

Common carotid intima-media thickness is predictive of all-cause and cardiovascular mortality in elderly community-dwelling people: Longitudinal Investigation for the Longevity and Aging in Hokkaido County (LILAC) study

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Several cohort studies have examined the association of carotid intima-media thickness (IMT) with the risk of stroke or myocardial infarction in apparently healthy persons. We investigated the predictive value of IMT of cardiovascular mortality in elderly community-dwelling people, beyond the prediction provided by age and MMSE, assessed by means of a multivariate Cox model. Carotid IMT and plaque were evaluated bilaterally with ultrasonography in 298 people older than 75 years (120 men and 178 women, average age: 79.6 years). The LILAC study started on July 25, 2000. Consultations were repeated every year. The follow-up ended on November 30, 2004. During the mean follow-up span of 1152 days, 30 subjects (21 men and nine women) died. Nine deaths were attributable to cardiovascular causes (myocardial infarction: two men and three women; stroke: two men and two women). The age- and MMSE-adjusted relative risk (RR) and 95% confidence interval (95% CI) of developing all-cause mortality was assessed. A 0.3 mm increase in left IMT was associated with a RR of predicted 1.647 (1.075-2.524), and a similar increase in right IMT with a RR of 3.327 (1.429-7.746). For cardiovascular mortality, the corresponding RR values were 2.351 (1.029-5.372) and 2.890 (1.059-7.891), respectively. Carotid IMT assessed by ultrasonography is positively associated with an increased risk of all-cause and cardiovascular death in elderly community-dwelling people.